



Tribhuvan University
Faculty of Humanities and Social Science

A Project Report on
Online Bus Reservation System

Submitted to
Department of Computer Application
Nepalaya College
Kalanki, Kathmandu

*In Partial fulfilment of the requirements of Bachelors in Computer
Application*

Submitted By
Sangam Sedai (6-2-577-19-2023)
Udesh Regmi (6-2-577-22-2023)

Under the Supervision of
Mr. Yuba Raj Devkota



Tribhuvan University
Faculty of Humanities and Social Science
Nepalaya College

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project proposal under my supervision by **Sangam Sedai & Udesch Regmi** entitled “**Online Bus Reservation System**” in partial fulfilment of the requirements for the degree of Bachelor in Computer Application is recommended for the final evaluation.

Yuba Raj Devkota
(SUPERVISOR)
Bachelor in Computer Application
Nepalaya College
Kalanki, Kathmandu



Tribhuvan University
Faculty of Humanities and Social Science
Nepalaya College

LETTER OF APPROVAL

This is to certify that this project proposed by “**Sangam Sedai & Udesb Regmi**” entitled “**Online Bus Reservation System**” in partial fulfilment of the requirements for the degree Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

.....
Yuba Raj Devkota,
Supervisor
Nepalaya College
Kalanki, Kathmandu

.....
Saroja Bhandari,
Coordinator
Nepalaya College
Kalanki, Kathmandu

.....
Internal Examiner

.....
External Examiner

ABSTRACT

The Online Bus Reservation System (OBRS) is a web-based platform developed to simplify and modernize bus ticket booking. It enables passengers to search routes, check available seats, reserve tickets, complete online payments, and receive electronic tickets instantly, eliminating the need for physical queues. The system provides a clean and user-friendly interface for customers, along with a dedicated admin panel for operators to manage buses, schedules, users, and bookings efficiently. Built using PHP, MySQL, and standard web technologies, the application ensures reliable data handling and smooth performance. It also incorporates essential system design components such as ER diagrams, data flow diagrams, and testing procedures to maintain accuracy and security. Overall, OBRS replaces traditional paper-based methods with a faster, more organized, and digital approach suitable for small and medium-scale transport businesses, improving efficiency, accessibility, and overall user satisfaction.

Keywords: *E-ticket, Online Bus Reservation, Reserve Ticket*

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project supervisor **Mr. Yuba Raj Devkota** for his continuous guidance, valuable suggestions, and constant encouragement throughout the completion of this project. His expertise, insightful feedback, and timely support played a crucial role in shaping the project and ensuring its successful completion.

We are also thankful to the Head of the Department and all the faculty members of Nepalaya College for providing the necessary academic support, facilities, and learning environment required for carrying out this project effectively. Their cooperation and guidance greatly contributed to enhancing our understanding of the subject matter.

We would like to extend our heartfelt thanks to our friends and family members for their moral support, patience, and encouragement during the entire project duration. Their motivation helped us remain focused and committed to our work.

Finally, we express our gratitude to Tribhuvan University for including *Project-I* in the BCA 4th Semester curriculum, which provided us with an opportunity to gain practical exposure to Software Engineering, Database Management Systems, and Scripting Languages. This project has significantly enhanced our technical knowledge and practical skills.

Regards,

Sangam Sedai (6-2-577-19-2023)

Udesh Regmi (6-2-577-22-2023)

January, 2026

TABLE OF CONTENTS

SUPERVISOR’S RECOMMENDATION	ii
LETTER OF APPROVAL	iii
ABSTRACT.....	iv
ACKNOWLEDGEMENT	v
TABLE OF CONTENTS.....	vi
LIST OF ABBREVIATIONS.....	viii
LIST OF FIGURES	ix
LIST OF TABLES.....	x
Chapter 1: Introduction	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scope and Limitations.....	2
1.5 Report Organization	4
Chapter 2: Background Study and Literature Review	5
2.1 Background Study	5
2.2 Literature Review.....	6
Chapter 3: System Analysis and Design.....	8
3.1 System Analysis.....	8
3.1.1 Requirement Analysis	8
3.1.2 Feasibility Analysis:	11
3.1.3 Data Modeling (ER Diagram)	13
3.1.4 Process Modeling (DFD).....	15
3.2 System Design	16
3.2.1 Architectural Design.....	16
3.2.2 Database Schema Design	17
3.2.3 Interface Design.....	18
3.2.4 Physical DFD	20
Chapter 4: Implementing & Testing	21
4.1 Implementation	21
4.1.1 Tools Used (CASE Tools, Programming Language, Database Platform)	21
4.1.2 Implementation Details of Modules	26

4.2	Testing.....	26
4.2.1	Test Case for Unit Testing.....	27
4.2.2	Test Case for System Testing	29
Chapter 5: Conclusion & Future Recommendations		35
5.1.	Lesson Learnt Outcome	35
5.2	Conclusion.....	35
5.3	Future Recommendations:	36

REFERENCES

APPENDICES

LIST OF ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ER	Entity-Relationship
HTML	Hyper Text Markup Language
JS	JavaScript
MySQL	My Structured Query Language (Database Management System)
OBRS	Online Bus Reservation System
PHP	Hypertext Preprocessor (Server-Side Scripting Language)
SQL	Structured Query Language
UI	User Interface
UX	User Experience

LIST OF FIGURES

Figure 3.1 Traditional Waterfall Model of OBRS	8
Figure 3.2 Use Case Diagram of OBRS	10
Figure 3.3 Schedule Feasibility of OBRS.....	13
Figure 3.4 ER Diagram of OBRS	14
Figure 3.5 Level 0 DFD of OBRS	15
Figure 3.6 Level 1 DFD of OBRS	15
Figure 3.7 Three Tier Architecture of OBRS	16
Figure 3.8 Schema Diagram of OBRS.....	17
Figure 3.9 Interface Design of OBRS.....	19
Figure 3.10 Physical DFD of OBRS.....	20
Figure 4.1 Frontend Code of OBRS	22
Figure 4.2 Backend Code of OBRS.....	24
Figure 4.3 SQL Code of OBRS	25

LIST OF TABLES

Table 3.1 Technical Feasibility Study of OBRS.....	11
Table 4.1 User Registration Test Case of OBRS	27
Table 4.2 User Login Test Case of OBRS	28
Table 4.3 Reservation Test Case of OBRS	28
Table 4.4 System Test Case of OBRS	29

Chapter 1: Introduction

1.1 Introduction

An Online Bus Reservation System is a web-based platform designed to automate and simplify the process of booking bus tickets, managing reservations, and handling passenger information. The OBRS enables passengers to book seats online in real time, reducing the need for physical reservations and long queues at ticket counters. With the increasing demand for digital services, people now expect fast, secure, and reliable solutions for travel planning. A bus reservation system provides a convenient interface where users can search for routes, view bus schedules, check seat availability, and complete bookings with online payment via simulated esewa or cash on counter. Administrators can manage bus details, users, reservations, seat allocations, and generate reports efficiently. By offering an easy-to-use and accessible platform, the Online Bus Reservation System enhances customer satisfaction, reduces manual workload, and improves operational efficiency for transport companies within Nepal.

Despite the growing adoption of digital platforms in Nepal's transportation sector, most existing online bus reservation solutions are designed for large commercial operators and remain inaccessible to small and medium-scale bus businesses that still rely on manual ticketing. This creates a clear research opportunity to develop a simple, affordable, and fully web-based reservation system that combines real-time seat booking, digital payment simulation, e-ticket generation, and complete admin control within a single open-source platform tailored for Nepal's domestic bus transport needs.

1.2 Problem Statement

Nepal's bus services still rely on manual ticket booking systems even though digitalization is rapidly growing. Bus operators use handwritten records and counter-based sales, which makes the process slow and prone to errors like overbooking, duplicate seat assignments, and lost records. These problems become even worse when demand is high. Passengers living in remote areas also face difficulties because they have no way to check seat availability, bus schedules, or route changes without physically traveling to a bus park. The Online Bus Reservation System (OBRS) solves these problems by providing a digital

platform where passengers can book seats online without visiting any counter. Seats are locked the moment a passenger selects them, and all booking data is stored securely in a MySQL database. Since OBRS is fully web-based, anyone with an internet connection can search for routes, check live seat availability, book tickets, and download e-tickets from anywhere in Nepal at any time.

1.3 Objectives

- To design and develop an Online Bus Reservation System as a web-based application using PHP, MySQL, HTML, CSS, and JavaScript.
- To provide functionalities including passenger registration, route searching, seat booking, online payment, and e-ticket generation, along with an administrative module to manage buses, routes, schedules, users, and bookings for complete reservation control.

1.4 Scope and Limitations

The Online Bus Reservation System (OBRS) is developed as a comprehensive, user-friendly web-based application aimed at modernizing the traditional bus ticket booking process in Nepal. It provides an efficient digital platform that allows passengers to book tickets conveniently from anywhere with internet access, while enabling bus operators and administrators to manage operations smoothly through a centralized interface. The system is built using modern web technologies to ensure responsiveness, security, and ease of use across devices. However, being an academic project, it has certain practical constraints and deliberate exclusions to keep the development focused, feasible, and aligned with the project timeline and resources.

Scope:

- The system provides a responsive web platform accessible on desktops, laptops, tablets, and mobile browsers, allowing passengers to search bus routes, check real-time seat availability, select preferred seats, and complete bookings without requiring any native application installation.
- Secure user registration and login system with password protection and session management is implemented, along with a user dashboard where passengers can view booking history, upcoming journeys, and cancel reservations when needed.

- An interactive seat selection interface displays real-time seat availability showing booked, available, and reserved seats, and an admin panel provides full control to manage bus routes, schedules, user accounts, and bookings with the ability to generate basic summary reports.
- The system includes a simulated online payment interface similar to eSewa for completing bookings, along with automatic generation and download of e-tickets containing unique booking ID, passenger information, and journey details.
- The system emphasizes paperless operations with a modular design that allows future addition of features such as SMS and email alerts, multilingual interface, and live payment gateway integration.

Limitations:

- The payment module is purely simulated and no real integration with live payment gateways such as eSewa is implemented, as it requires official API credentials, security compliance, and legal approvals that are beyond the scope of this academic project.
- Real-time bus tracking using GPS, dynamic fare pricing, and advanced third-party integrations such as Google Maps or hotel booking services are not included in the current version of the system.
- The system is designed exclusively for domestic intra-Nepal bus routes and full functionality depends on a stable internet connection, as no offline mode is available for users in areas with poor or no connectivity.

1.5 Report Organization

This report is organized into several structured chapters to provide a clear understanding of the Online Bus Reservation System its purpose, design, implementation, and overall functionality. Each chapter focuses on a specific aspect of the system so that readers can grasp both the technical and conceptual details effectively.

- **Chapter 1: Introduction**

This chapter provides an overview of the Online Bus Reservation System, including the background of the tours and travelling industry specially bus & the motivation behind developing the system, and the problems it aims to solve. It explains the scope, objectives, and significance of the project.

- **Chapter 2: Literature Review / Background Study**

This section presents an analysis of existing reservation systems, booking platforms, and bus portals. It identifies their limitations and highlights the areas where OBRS, improves upon current solutions, particularly in user convenience, automation, and integrated travel services.

- **Chapter 3: System Analysis and Design**

This chapter includes requirement analysis, feasibility study, and problem definition. Both functional and non-functional requirements are documented, along with use-case descriptions that illustrate how users interact with the system. System workflows and design strategies for the Online Bus Reservation platform are also discussed.

- **Chapter 4: Implementation and Testing**

This chapter explains the system architecture, database schema, Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and the overall development methodology. It describes how different components—such as bus reservation, bus adding and update and how data flows throughout the Online Bus Reservation System. Testing procedures and results are also included.

- **Chapter 5: Conclusion and Future Recommendations**

This section summarizes the technologies used, the development tools, coding structure, and the module-wise implementation of the Online Bus Reservation System. Screenshots and interface descriptions are provided to demonstrate system functionality. The chapter concludes with recommendations for future enhancements and additional features.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Bus transportation remains one of the most essential and commonly used means of travel in Nepal, serving as a primary link between major urban centers such as Kathmandu, Pokhara, Chitwan, Biratnagar, and Lumbini, as well as providing connectivity to remote and rural districts. Despite its importance in the daily lives of millions of passengers, the existing ticketing system is largely traditional and manual, requiring individuals to physically visit ticket counters to reserve seats. This outdated approach leads to several operational challenges, including long waiting lines, limited seat availability, lack of pricing transparency, ticket fraud, and frequent human errors in maintaining records [4][8]. The situation becomes more problematic during peak travel periods such as Dashain, Tihar, and other national holidays, when the demand for tickets significantly increases while the manual system fails to efficiently handle the surge [3][6]. Additionally, small and medium-scale bus operators face difficulties in managing routes, schedules, and passenger information due to the absence of a structured digital system, often resulting in issues like overbooking and inefficient service delivery [7][8].

With the expansion of internet accessibility and the growing adoption of digital payment platforms such as eSewa, Khalti, and online banking in Nepal, there is a strong technological base for implementing effective online ticketing systems. Existing platforms like BusSewa, Hamro Bus, and TicketSewa demonstrate that digital bus reservation is both practical and widely accepted among users [1][5][9]. However, many small and medium transport operators still lack access to customized digital solutions and continue to rely on manual processes [4][6]. To address these limitations, the Online Bus Reservation System (OBRS) has been developed as a web-based solution utilizing technologies such as PHP, MySQL, HTML, CSS, and JavaScript. This system enables passengers to search available routes, check real-time seat availability, select preferred seats, perform secure online payments, and receive instant electronic tickets from any location at any time. At the same time, it provides an administrative interface for operators to efficiently manage buses, routes, schedules, bookings, and reports, thereby eliminating the need for manual paperwork and improving overall operational efficiency [2][3][4].

2.2 Literature Review

Several studies and existing systems have been reviewed to understand how online bus reservation systems are built, what technologies are commonly used, and what challenges exist in their development. BusSewa [1] is Nepal's first real-time online bus ticket booking platform developed by Diyalo Technologies Pvt. Ltd., which allows passengers to view live seat availability and book tickets at transparent prices with no additional booking charges. This platform demonstrated that a centralized digital ticketing solution is both feasible and necessary in Nepal's transportation sector and serves as a strong practical reference for understanding what passengers in Nepal expect from an online reservation system. Christian [2] developed a bus ticket reservation system using PHP and MySQLi, featuring a seat inventory mechanism that effectively prevents double-booking and allows passengers to reserve seats instantly and print their tickets, showing that even a simple PHP/MySQLi-based system can solve critical seat conflict problems in online reservations and improve overall reliability. Ghara et al. [3] developed an online bus ticket reservation system using PHP, MySQL, HTML, CSS, JavaScript, and AJAX, demonstrating that AJAX-based design improves system responsiveness and overall user experience. However, a common limitation noted in their study was the absence of a proper admin dashboard for bus operators, which often leads to data inconsistencies over time. Ghimire et al. [4] confirmed through an academic-level Nepali project that PHP and MySQL applications can effectively handle route management, seat booking, and admin operations at a small to medium scale, further supporting the technical approach used in OBRS.

Hamro Bus [5] established that private-sector digital platforms can play a key role in modernizing Nepal's bus travel industry by offering transparent pricing, best seat guarantees, and quality customer service, reinforcing the demand for reliable online booking systems in Nepal. Oloyede et al. [6] developed a web-based reservation system using XHTML, PHP, and SQL following the Structured Systems Analysis and Design Methodology (SSADM), confirming that automating manual ticketing processes significantly reduces long queues and human errors, though their system lacked mobile-friendliness and real-time seat tracking, identifying clear areas for improvement. Saputra et al. [7] addressed many of these gaps by building a prototyping-based web reservation system that reported up to 50% improvement in booking efficiency, reduced data entry errors, and high user satisfaction, supporting the prototyping model as an effective

development approach for bus reservation systems. Singh [8] developed a similar web-based Bus Ticket Booking System using PHP, MySQL, HTML, CSS, and JavaScript, highlighting that manual ticketing causes delays and errors and that a computerized system reduces workload and improves customer satisfaction, though the system lacked online payment integration and was limited to small-scale use. TicketSewa [9] demonstrated that a professional, end-to-end digital ticketing solution for Nepal's bus industry is fully achievable, offering complete e-ticketing, online payment, and real-time booking confirmation as a successful commercial reference for what OBRS aims to provide at an academic level. From the above review, it is clear that web-based bus reservation systems using PHP and MySQL are widely adopted and proven effective. Most existing systems, however, focus heavily on passenger-facing features while admin-side management remains underdeveloped. The OBRS project directly addresses these gaps by delivering a complete system with both a passenger interface and a comprehensive admin panel, real-time seat availability, secure booking workflow, and instant e-ticket generation.

Chapter 3: System Analysis and Design

3.1 System Analysis

System analysis is the process of studying and understanding a system so it can be developed in an organized way. For OBRS, the requirements are clear and stable, so the project follows the waterfall model. Each phase is completed before moving to the next, which helps maintain a proper flow and ensures the system is built in a systematic and reliable manner.

In Waterfall, every phase relies upon the results of the previous phase. Prior to moving and going for the next phase, a proper review of the previous phase is conducted, and this helps keep everything consistent and clear in the whole process of development. As this proposed system has fixed specifications and does not need many adjustments and additions, the Waterfall model is the best approach for developing software related to the OBRS.

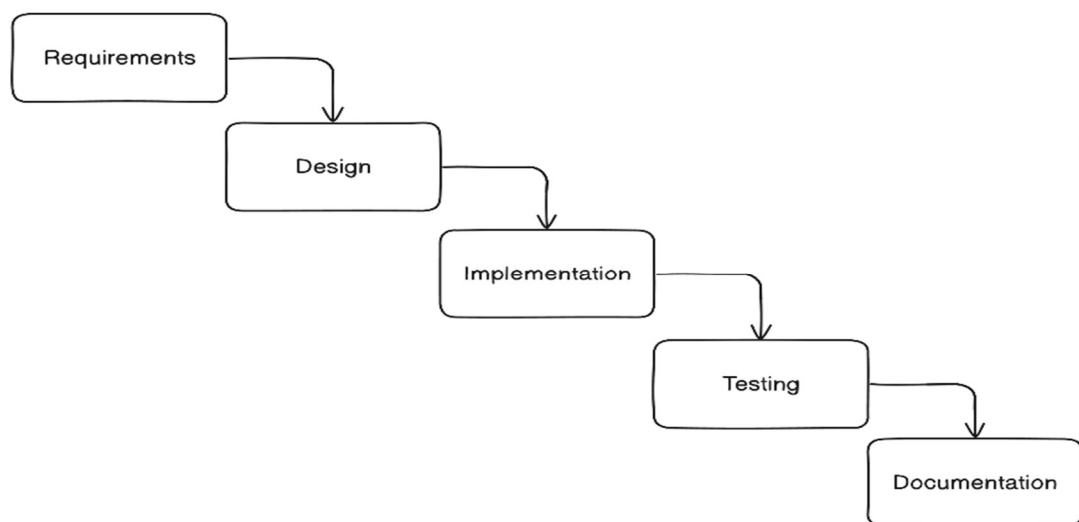


Figure 3.1 Traditional Waterfall Model of OBRS

3.1.1 Requirement Analysis

The system requires the following functional and non-functional requirements

- **Functional Requirement:**

The functional requirements define the core operations and features of the Online Bus Reservation System (OBRS). These describe how the system behaves and what services it provides to users and administrators.

- **User Registration and Authentication:** The system allows new users to create an account by providing necessary details such as username, email, and password. Registered users can securely log in using their credentials. The system validates user information during login and ensures that only authenticated users can access their respective dashboards.
- **Reservation Management:** The system enables users to search available bus routes based on source, destination, and travel date. Users can view seat availability, select seats using an interactive layout, and proceed with booking. After successful reservation, the system generates an e-ticket which can be viewed or downloaded. Users can also manage their bookings through their dashboard.
- **Admin Management:** The system provides an administrative panel for managing overall operations. Admin users can add, update, or delete bus details, routes, and schedules. They can monitor bookings, manage registered users, and oversee system activities. This ensures proper control and smooth functioning of the reservation system.

A use case is a methodology used in system analysis to identify, clarify and organize system requirements. Use case diagram for the proposed system is given below:

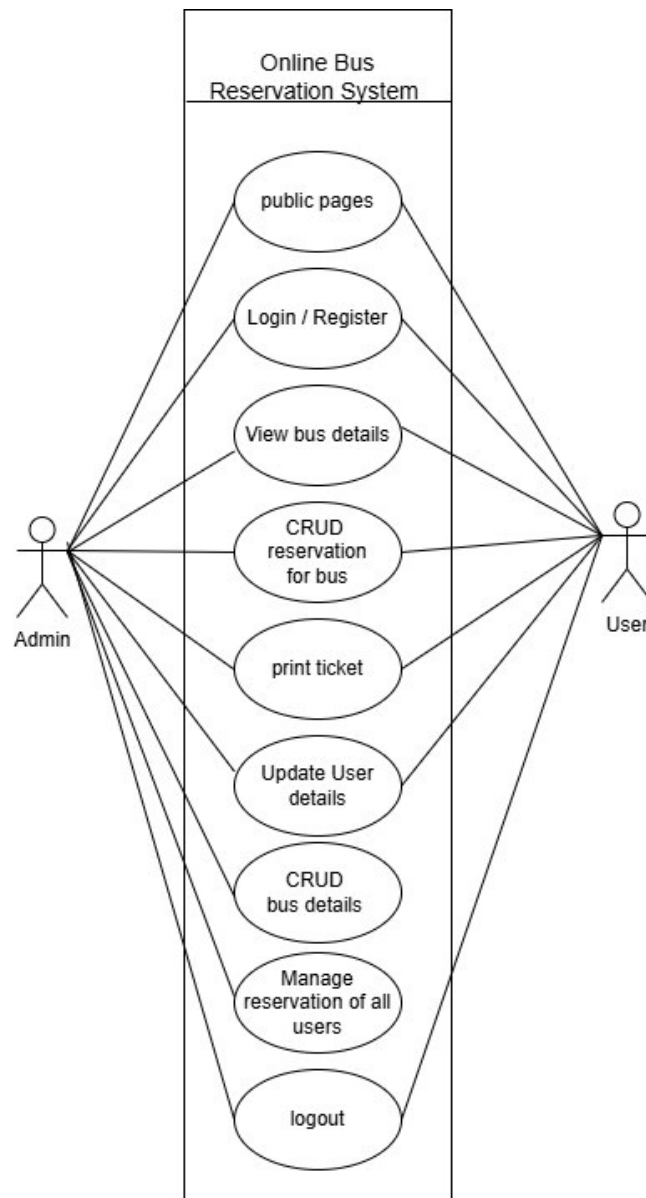


Figure 3.2 Use Case Diagram of OBRS

- **Non-Functional Requirement:**

The non-functional requirements define the quality attributes and performance standards of the Online Bus Reservation System (OBRS). These ensure that the system operates efficiently, securely, and provides a good user experience.

- **Security:** The system ensures that only authorized users can access its features. User authentication is implemented through login credentials, and access control is managed based on user roles. Each user is allowed to access only their respective

dashboard, while administrative privileges are restricted to admin users. This prevents unauthorized access and protects sensitive data.

- **Performance:** The OBRS is designed to deliver smooth and efficient performance. The system is optimized using standard web development practices to ensure fast loading times and responsive interactions. Under a stable internet connection, users can search routes, book tickets, and access services without noticeable delays.
- **Reliability:** The system is built to provide consistent and dependable service. When deployed in a production environment with a stable internet connection, it remains accessible to users at all times. Proper handling of user requests and system processes ensures minimal errors and continuous operation.
- **User Friendliness:** The system provides a simple and interactive user interface. Features such as seat selection layout, clear navigation, and separate dashboards for users and administrators make the system easy to use. The design focuses on simplicity so that users can perform tasks like booking tickets and managing data without confusion.

3.1.2 Feasibility Analysis:

A feasibility study assesses the practicality of the proposed Online Based Bus Reservation System (OBRS) by evaluating technical, operational, economic, and schedule aspects prior to development.

- **Technical Feasibility:**

The system is technically viable using free, open-source technologies: HTML, CSS, and JavaScript for the frontend, PHP for server-side scripting, and MySQL for database management. These tools support full CRUD operations without external APIs.

Table 3.1 Technical Feasibility Study of OBRS

Technological Requirement	Hardware Requirement	Software Requirement
HTML	Laptop / PC	VS Code
CSS	Minimum 4GB RAM	Figma
JS	Keyboard	XAMPP Server
MYSQL	Mouse	MySQL /phpMyAdmin
PHP		MS Word
		Web Browser

- **Operational Feasibility:**

The system is operationally feasible, designed for users with varying digital literacy in Nepal. Passengers can search routes, select seats, and generate e-tickets through an intuitive web interface, while administrators manage buses, routes, and bookings via a secure dashboard. Minimal training is needed, with a short manual for users and basic guidance for admins on CRUD operations. Its web-based design allows deployment on affordable local hosting, ensuring easy maintenance and updates. High user acceptance is expected due to the growing use of online ticketing among Nepali commuters and tourists.

- **Economic Feasibility:**

OBRs is economically viable, relying on free open-source tools with minimal costs for domain registration (NPR 1,000–2,000/year) and shared hosting (NPR 3,000–8,000/year). Hardware requirements are met by existing laptops, and maintenance costs remain low, scaling with user demand. The system reduces manual labor, prevents overbooking, and enables revenue from premium features. ROI is quick, especially for medium-sized operators, making OBRs highly cost-effective in Nepal.

- **Schedule Feasibility:**

Schedule feasibility is critical for project success, as delays could render the system outdated. The project timeline spans four months, following the Waterfall methodology with defined phases: requirements gathering (25 days), design (27 days), implementation (45 days), testing (23 days), and documentation/deployment (120 days).

A schedule chart was prepared to visualize and monitor progress, ensuring milestones are met within the BCA semester constraints. Tasks were allocated considering the team's capacity (two members), with buffers for unforeseen issues like testing iterations. This structured approach confirms that the project can be completed on time.

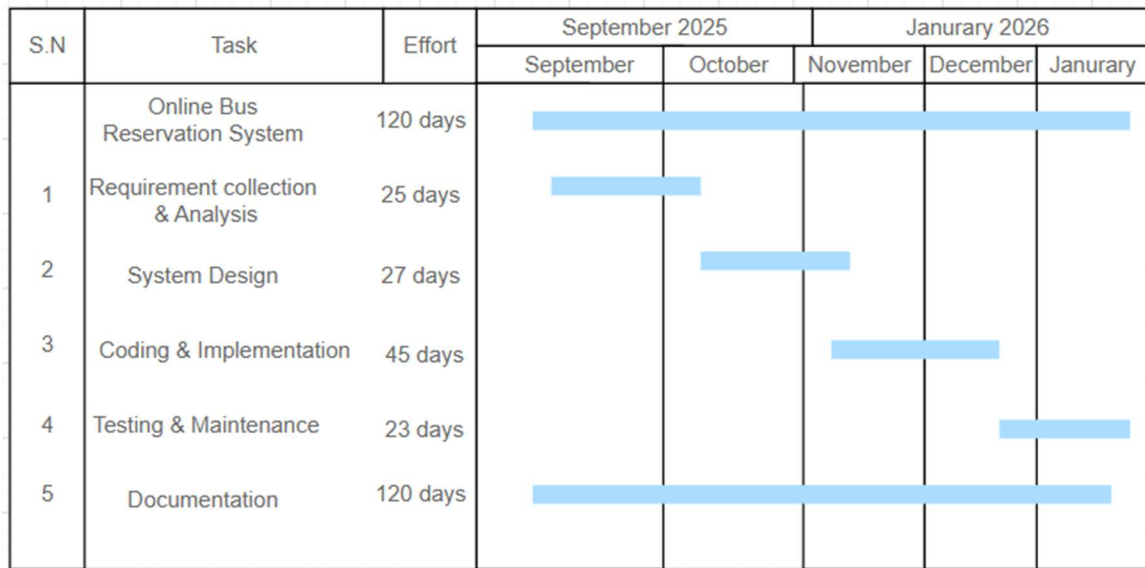


Figure 3.3 Schedule Feasibility of OBRS

3.1.3 Data Modeling (ER Diagram)

An Entity-Relationship Diagram (ER Diagram) is a visual representation that illustrates the logical structure of the database by defining entities, their attributes, and relationships. ER diagrams are constructed using three core components: entities (real-world objects), attributes (properties of entities), and relationships (associations between entities). In OBRS, the ER model ensures data consistency and directly maps to the MySQL database schema. Key entities include User (passengers and admins), Bus, reservation. Relationships such as "users makes reservation on resevations" and "reservation reserves Bus" and if user is admin "User manages bus".

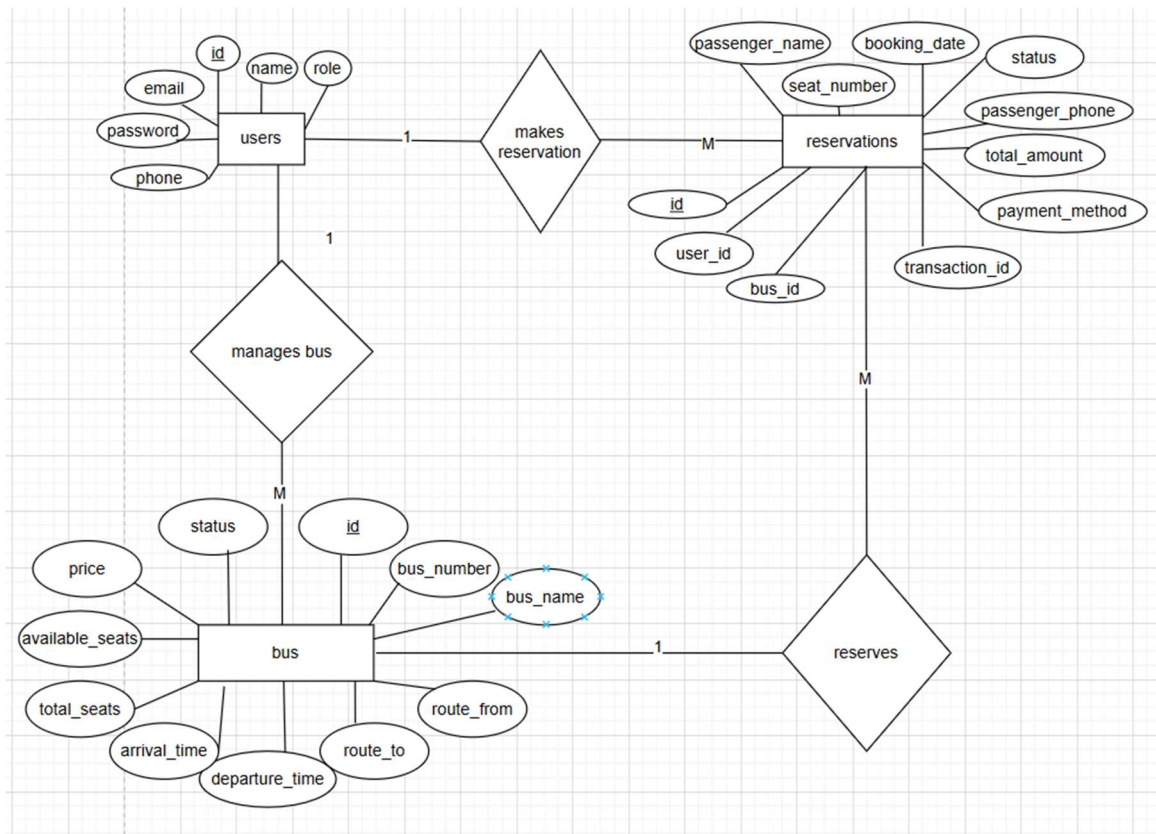


Figure 3.4 ER Diagram of OBRS

3.1.4 Process Modeling (DFD)

A Data Flow Diagram (DFD) graphically depicts the flow of data within the system, highlighting processes, data stores, external entities, and data movements. It serves as a preliminary overview for system elaboration, focusing on what data is input/output, sources/destinations, and storage without detailing timing or sequence. In OBRS, DFDs are limited to Level 0 (Context) & Level 1 for clarity and validity.

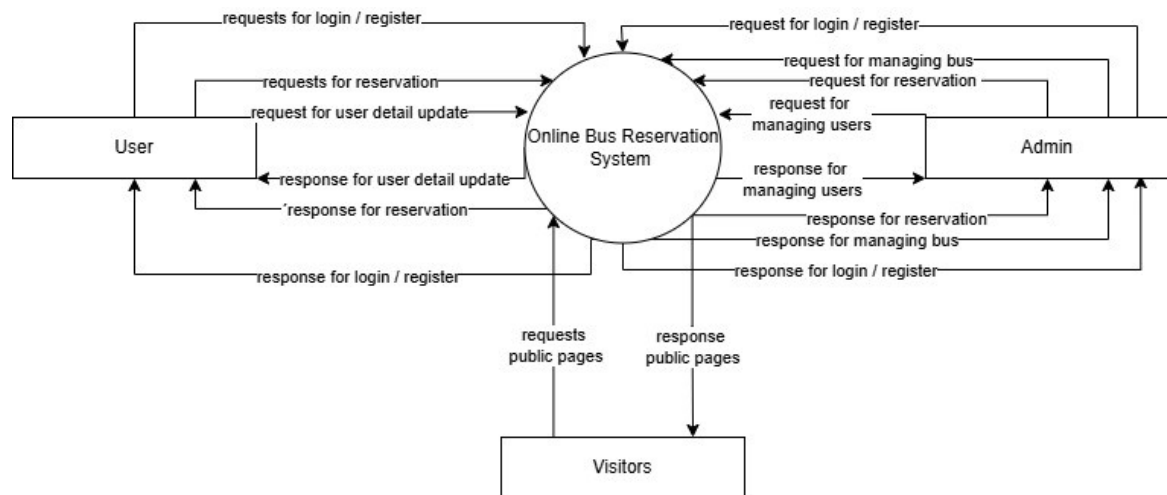


Figure 3.5 Level 0 DFD of OBRS

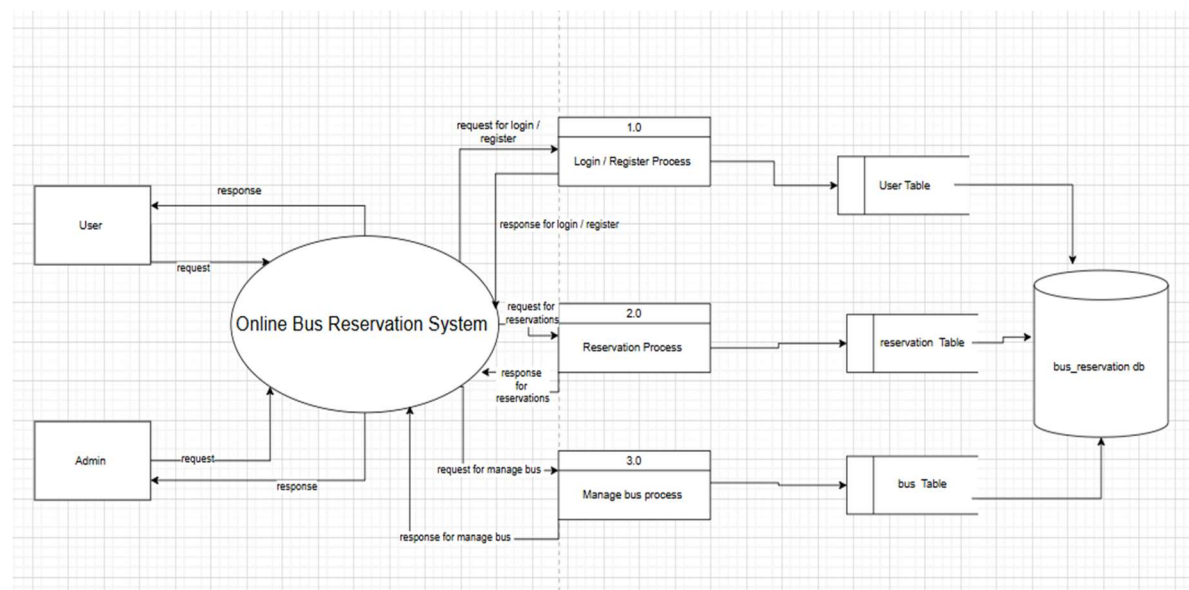


Figure 3.6 Level 1 DFD of OBRS

3.2 System Design

System design involves defining the architecture, components, modules, and interfaces to fulfill user requirements effectively.

3.2.1 Architectural Design

The system adopts a three-tier architecture to ensure modularity, scalability, and maintainability, suitable for web-based applications in resource-constrained environments like Nepal.

- **Client Layer** (Presentation Tier): The Client layer of OBRS handles user interface via browser, rendering dynamic pages with HTML, CSS, JavaScript, with responsiveness.
- **Server Layer** (Business Logic Tier): The PHP scripts process requests, validate data, perform CRUD operations, and interact with the database for the OBRS.
- **Database Layer (Data Tier)**: MySQL stores persistent data independently, ensuring security and efficiency for OBRS.

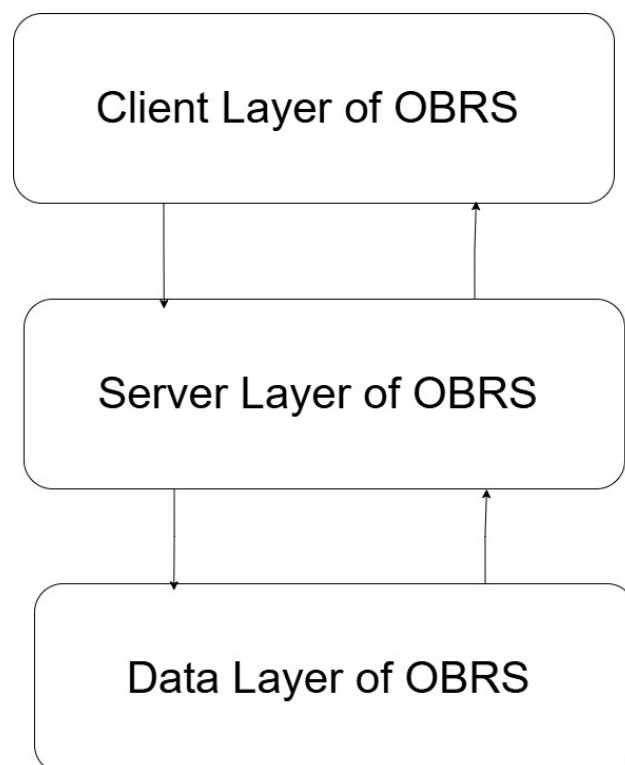


Figure 3.7 Three Tier Architecture of OBRS

3.2.2 Database Schema Design

A database schema is the skeleton structure that represents the logical view of the entire database. It defines how the data is organized and how the relations among them are associated. It formulates all the constraints that are to be applied on the data. A database schema defines its entities and the relationship among them. The database schema defines the structure of the MySQL database used in the Online Bus Reservation System. It consists of the following main tables:

- **Users Table:** Stores information about users and administrators, including user ID, name, login credentials, and assigned roles.
- **Reservations Table:** Contains details of reservation of a particular user for a particular bus.
- **Buses Table:** Contains details of bus details which can only be created by the user whose role is admin.

The schema maintains data integrity through the use of primary keys and foreign keys, ensuring accurate relationships between tables while preventing duplication and inconsistency in records.

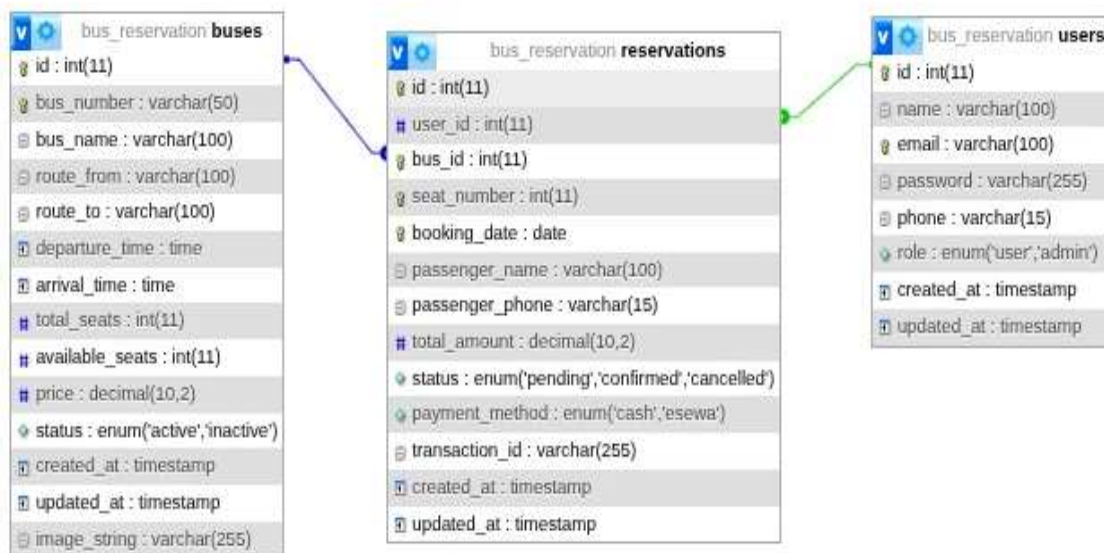
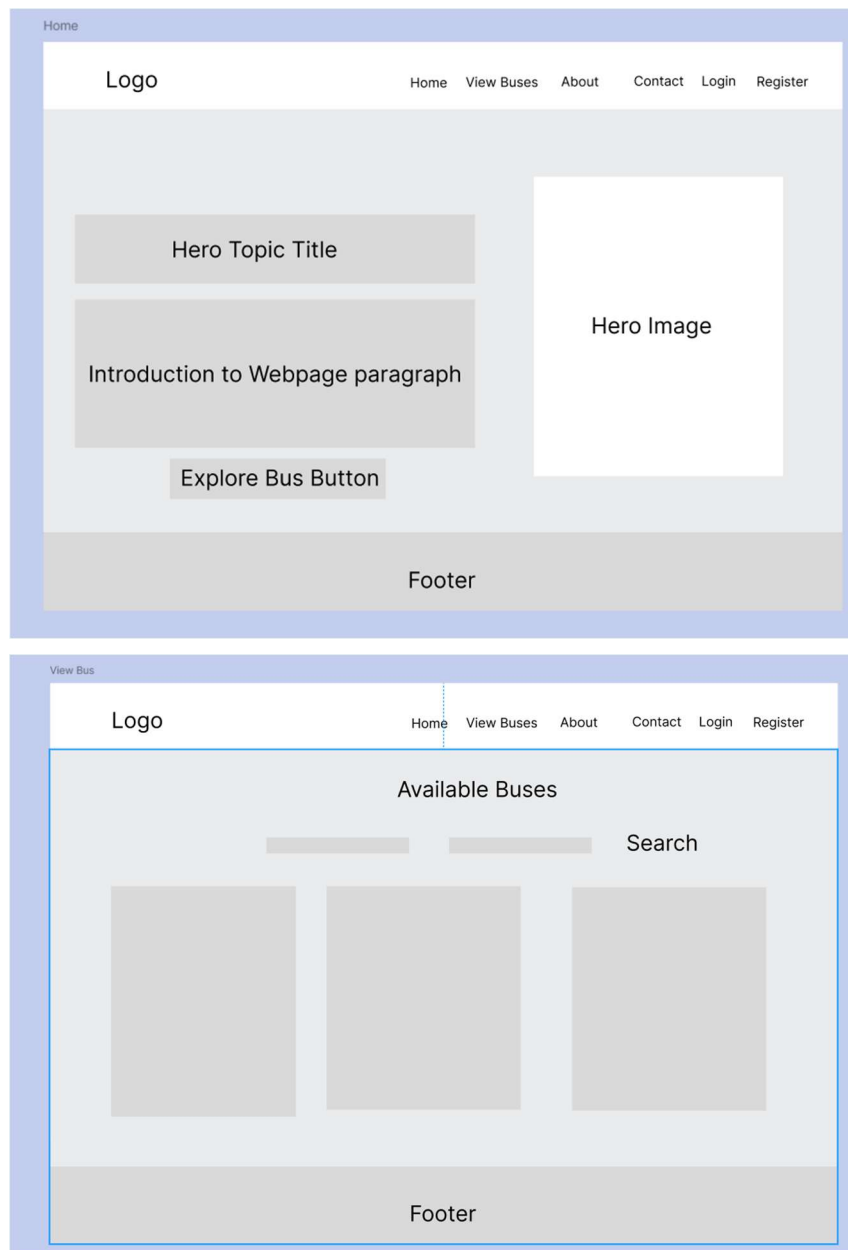


Figure 3.8 Schema Diagram of OBRS

3.2.3 Interface Design

The system interface is designed with usability as a priority, featuring clearly separated workflows for users and administrators. The user interface enables passengers to search buses, view schedules, select seats, and complete reservations efficiently. The administrative interface provides authorized personnel with tools to create, view, update, and delete bus records, routes, and schedules. This role-based design ensures smooth reservation handling while maintaining secure and efficient system management.



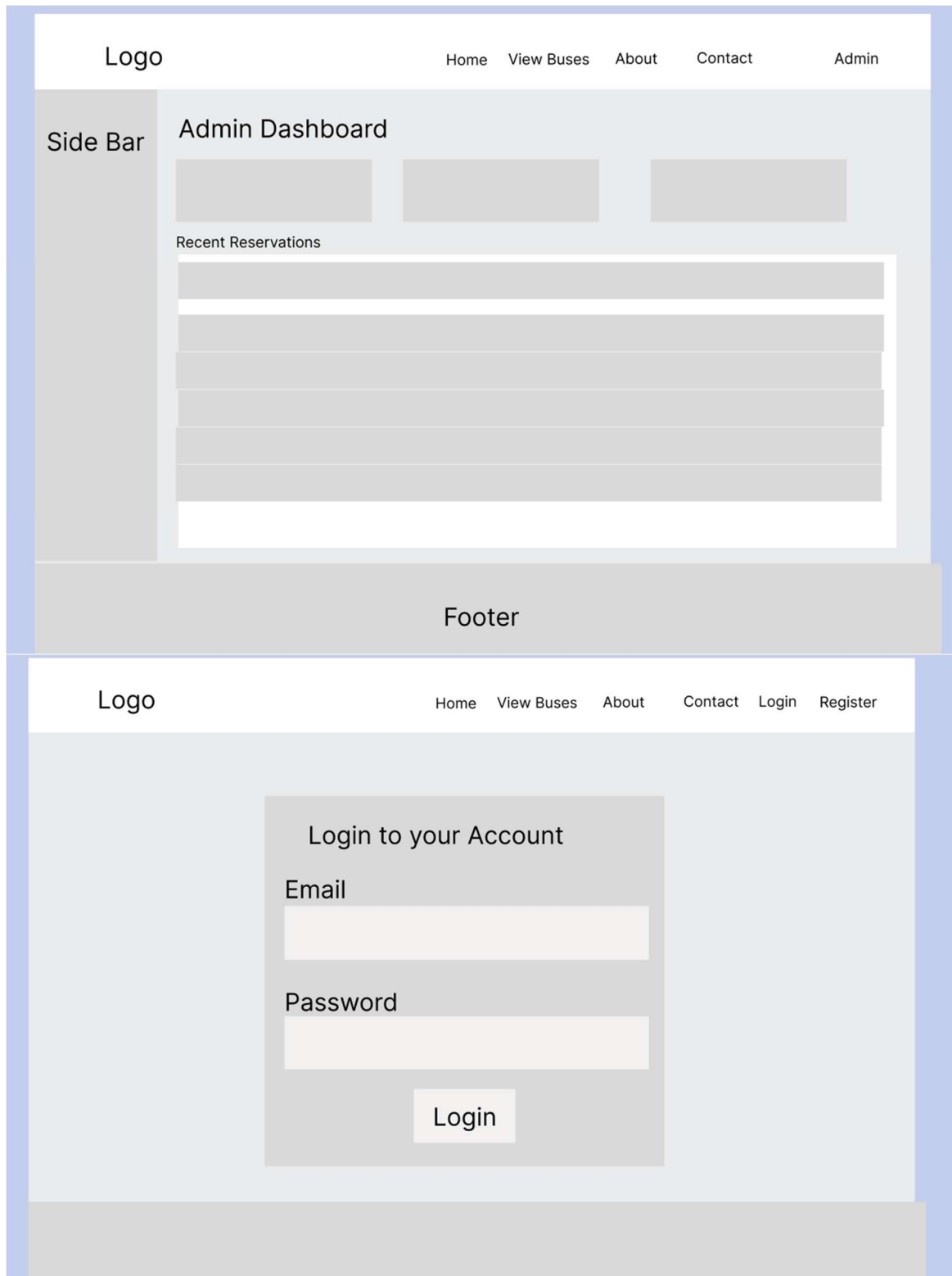


Figure 3.9 Interface Design of OBRS

3.2.4 Physical DFD

The Physical Data Flow Diagram of the Online Bus Reservation system represents how data flows through actual system components including the web browser, Apache server, PHP application layer, and MySQL database. Users interact through the web interface by entering details such as source, destination, travel date, and login credentials, which are transmitted via HTTP requests to the PHP backend for processing. Login and registration requests are validated against the Users Table, while bus search queries retrieve matching routes, schedules, and seat availability from the Buses Table. Upon seat selection and booking confirmation, the PHP backend inserts a reservation record into the Reservations Table and generates an e-ticket for the user. The Admin manages buses, routes, and reservations through a role-based secured panel. All operations ensure continuous real-time data exchange between the frontend, PHP backend, and centralized MySQL database.

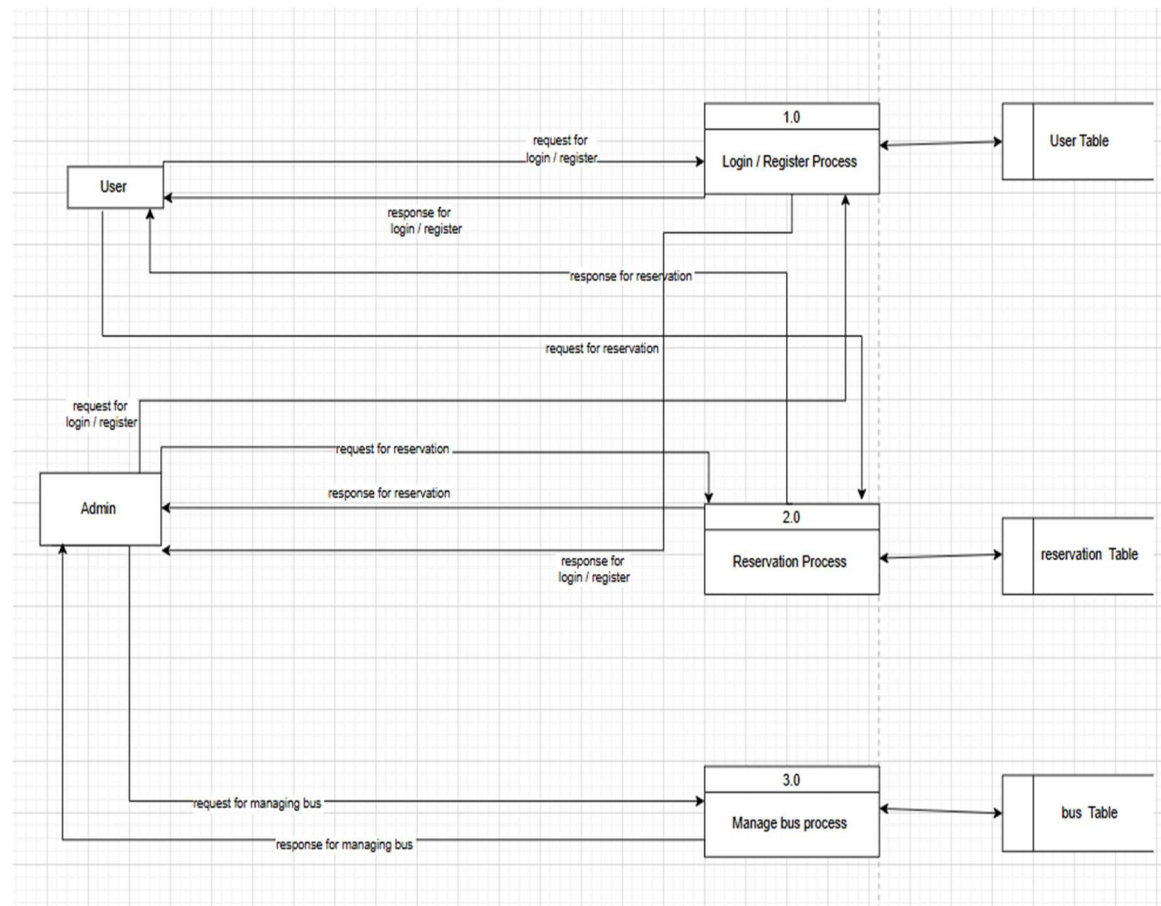


Figure 3.10 Physical DFD of OBRS

Chapter 4: Implementing & Testing

4.1 Implementation

Implementation is the phase in which the system design is translated into a fully functional Online Bus Reservation System. During this stage, all planned modules are developed, configured, and integrated according to the approved design specifications. Core components such as user registration, bus and route management, seat selection, reservation processing, and e-ticket generation are implemented to ensure a smooth and reliable booking workflow.

The implementation process emphasizes the development of a responsive and user-friendly interface for passengers, allowing them to search routes, check seat availability, and complete reservations efficiently. At the same time, a secure administrative panel is built to enable operators to create, update, and delete bus records, schedules, and booking details through CRUD operations. Strong focus is placed on backend security, accurate data handling, and seamless interaction between the client and admin modules. Overall, this phase ensures that the system operates reliably, supports real-time updates, and delivers an efficient online reservation experience for both users and transport operators.

4.1.1 Tools Used (CASE Tools, Programming Language, Database Platform)

This section describes the tools and technologies used in developing both the frontend and backend of the Online Bus Reservation System.

Frontend

The frontend of Online Bus Reservation System is developed using standard web technologies to create an interactive and user-friendly interface.

```

<footer class="site-footer">
  <div class="footer-container">
    <div class="footer-brand">
      <h3 class="footer-title"><?php echo SITE_NAME; ?></h3>
      <p class="footer-tagline">Reliable travel. Simple booking.</p>
    </div>

    <div class="footer-links">
      <a href="<?php echo BASE_URL . '/pages/public/home.php' ?>">Home</a>
      <a href="<?php echo BASE_URL . '/pages/public/about.php' ?>">About</a>
      <a href="<?php echo BASE_URL . '/pages/public/contact.php' ?>">Contact</a>
      <a href="<?php echo BASE_URL . '/pages/public/viewbus.php' ?>">View Bus</a>
      <a href="<?php echo BASE_URL . '/pages/public/login.php' ?>">Login</a>
      <a href="<?php echo BASE_URL . '/pages/public/register.php' ?>">Register</a>
    </div>

    <div class="footer-legal">
      <p>&copy; <?php echo date("Y"); ?> <?php echo SITE_NAME; ?>. All rights reserved.</p>
      <p>Developed By <span><a href="https://www.linkedin.com/in/udesh-regmi/">Udesh Regmi </a></span></p>
    </div>
  </div>
</footer>

```

```

.site-footer {
  width: 100%;
  margin-top: var(--space-8);
  border-top: 1px solid var(--border);
  background: var(--black);
  color: var(--white);
}

.footer-container {
  max-width: 1200px;
  margin: 0 auto;
  padding: var(--space-6) var(--space-4);
  display: grid;
  gap: var(--space-4);
}

const forms = document.querySelectorAll('form');
forms.forEach(form => {
  form.addEventListener('submit', function(e) {
    const password = form.querySelector('input[name="password"]');
    const confirmPassword = form.querySelector('input[name="confirm_password"]');

    if (password && confirmPassword) {
      if (password.value !== confirmPassword.value) {
        e.preventDefault();
        showAlert('Passwords do not match!', 'danger');
        return false;
      }

      if (password.value.length < 6) {
        e.preventDefault();
        showAlert('Password must be at least 6 characters long!', 'danger');
        return false;
      }
    }
  });
});

```

Figure 4.1 Frontend Code of OBRS

HTML (Hypertext Markup Language)

HTML is used to structure all the pages of the Online Bus Reservation System (OBRS). Semantic tags such as <header>, <nav>, <main>, <footer>, <aside>, <section>, and <table> are used throughout the system to ensure proper page organization, accessibility, and clean code structure. Each module of OBRS including the public home page, login and registration forms, bus seat layout, reservation pages, and the admin dashboard are built using these semantic elements to maintain a well-structured and readable document hierarchy.

CSS (Cascading Style Sheets)

CSS is used to style the entire OBRS platform to make it visually appealing and consistent. Global design properties such as primary colors, font sizes, and spacing are defined using CSS :root custom variables so that the styling of the whole system can be managed from a single place. Class selectors are used throughout all pages to apply styles efficiently and avoid code repetition. CSS also handles the responsive layout of the system, ensuring that the public home page, seat selection page, and admin panel display correctly on desktops, tablets, and mobile devices.

JavaScript

JavaScript is used to add interactivity and dynamic behavior to OBRS. Client-side form validation is implemented using pattern attributes and regular expressions (regex) to verify that user inputs such as phone numbers, email addresses, and passwords are in the correct format before being sent to the server. JavaScript is also used for dynamic seat selection on the bus layout, filtering available routes, and handling asynchronous requests using the fetch() API to communicate with the backend without requiring full page reloads, providing a smoother experience for passengers.

Backend

The backend of OBRS is responsible for processing all user requests, managing reservations, handling user authentication, and ensuring secure communication between the frontend interface and the database.

```

requireLogin();

$database = new Database();
$db = $database->getConnection();
$user = new User($db);

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $action = $_POST['action'] ?? '';

    if ($action === 'update_profile') {
        $name = sanitize($_POST['name']);
        $email = sanitize($_POST['email']);
        $phone = sanitize($_POST['phone']);

        if ($user->update($_SESSION['user_id'], $name, $email, $phone)) {
            $_SESSION['user_name'] = $name;
            $_SESSION['user_email'] = $email;
            showAlert('Profile updated successfully!', 'success');
        } else {
            showAlert('Failed to update profile', 'danger');
        }

        redirect(BASE_URL . '/pages/user/profile.php');
        exit();
    }
}

```

```

<?php
require_once __DIR__ . '/../config/database.php';
require_once __DIR__ . '/../config/constants.php';
require_once __DIR__ . '/../includes/session.php';
require_once __DIR__ . '/../includes/functions.php';
require_once __DIR__ . '/../models/Bus.php';

// Check if user is admin
if (!isLoggedIn() || !isAdmin()) {
    showAlert('Unauthorized access', 'danger');
    redirect(BASE_URL . '/pages/public/home.php');
    exit();
}

```

Figure 4.2 Backend Code of OBRS

PHP

PHP is used as the server-side scripting language for OBRS. It manages user registration, login authentication, session handling, booking processing, and admin operations such as managing buses, routes, schedules, and user accounts. All database interactions in OBRS are performed using PHP Data Objects (PDO) with prepared statements, which protect the system against SQL injection attacks and ensure all data is handled securely and efficiently.

MySQL Database

MySQL is used to store all system data including user information, bus details, route configurations, seat availability, and reservation records. Tables are properly normalized and relationships between entities such as users, buses, routes, and reservations are defined using primary keys and foreign keys to maintain data integrity. Queries are optimized using indexed fields to ensure fast and reliable data retrieval even as the number of bookings grows over time.

```

1 CREATE DATABASE IF NOT EXISTS bus_reservation CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
2 USE bus_reservation;
3
4 -- Users Table
5 CREATE TABLE users (
6     id INT AUTO_INCREMENT PRIMARY KEY,
7     name VARCHAR(100) NOT NULL,
8     email VARCHAR(100) UNIQUE NOT NULL,
9     password VARCHAR(255) NOT NULL,
10    phone VARCHAR(15),
11    role ENUM('user', 'admin') DEFAULT 'user',
12    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
13    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
14    INDEX idx_email (email),
15    INDEX idx_role (role)
16 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
17
18 -- Buses Table
19 CREATE TABLE buses (
20     id INT AUTO_INCREMENT PRIMARY KEY,
21     bus_number VARCHAR(50) UNIQUE NOT NULL,
22     bus_name VARCHAR(100) NOT NULL,
23     route_from VARCHAR(100) NOT NULL,
24     route_to VARCHAR(100) NOT NULL,
25     departure_time TIME NOT NULL,
26     arrival_time TIME NOT NULL,
27     total_seats INT NOT NULL,
28     available_seats INT NOT NULL,
29     price DECIMAL(10,2) NOT NULL,
30     status ENUM('active', 'inactive') DEFAULT 'active',
31     image_string VARCHAR(255) DEFAULT NULL,
32     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
33     updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
34     INDEX idx_status (status),
35     INDEX idx_route (route_from, route_to),
36     INDEX idx_bus_number (bus_number)
37 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
38
39 -- Reservations Table
40 CREATE TABLE reservations (
41     id INT AUTO_INCREMENT PRIMARY KEY,
42     user_id INT NOT NULL,

```

Figure 4.3 SQL Code of OBRS

Development Tools

- **VS Code:** Used as the primary code editor for writing and organizing all HTML, CSS, JavaScript, and PHP project files.
- **XAMPP:** Used to set up a local Apache and MySQL server environment for development and testing of the system before deployment.
- **phpMyAdmin:** Used to design, manage, and view the OBRS database tables and relationships visually during development.
- **GitHub:** Used for version control, code backup, and tracking changes made throughout the development of the project.
- **Brave Browser:** Used to render, test, and debug the OBRS web pages during the development phase.
- **Figma:** Used for designing and prototyping the user interface of OBRS before the actual implementation began.

4.1.2 Implementation Details of Modules

The system comprises two primary modules Client Module (for passengers) and Admin Module (for operators). All modules share a common database and authentication system.

User Module:

- **Registration and Login:** Users register with name, email, phone, and password. Passwords are hashed using PHP's `password_hash()` for security. Login uses sessions (`$_SESSION`) to maintain authenticated state.
- **Bus Search and Booking:** Users search buses by destination. PHP queries MySQL (`SELECT * FROM buses JOIN routes ... WHERE ...`) to display available buses. Seat selection uses a grid layout; booked seats are fetched dynamically to prevent double-booking.
- **Payment Simulation and E-Ticket:** Payment is simulated (no live gateway). Upon "payment," a booking record is inserted, and an e-ticket PDF is generated using a simple HTML-to-PDF approach.
- **Booking History and Cancellation:** Users view past bookings and cancel (with status update to "cancelled").

Admin Module:

- **Dashboard:** Secure login (role = 'admin'). Overview shows total bookings, revenue summary, and recent activity.
- **CRUD Operations:**
 - Add/Edit/Delete Buses and Routes via forms.
 - View/Manage Reservation (view, cancel, delete).
 - User management (view users).

All operations use prepared statements PDO to prevent SQL injection.

4.2 Testing

The testing phase is a critical component of the software development lifecycle for the Online Based Bus Reservation System (OBRS), ensuring the application's reliability, functionality, security, and user satisfaction before deployment. Given the system's web-based nature and its integration with Nepal's transportation context, testing was conducted systematically to identify and rectify defects, validate requirements, and confirm performance under simulated real-world conditions. This encompassed both unit testing, which isolates and verifies individual components, and system testing, which evaluates the

integrated application as a whole. Manual testing techniques were primarily employed, supplemented by tools like browser developer consoles and Postman for API-like endpoint checks, as the project focuses on core PHP functionality without dedicated automated test frameworks. A total of 50 test cases were executed across modules, achieving 98% pass rate after iterations, with defects logged and resolved to minimize risks such as overbooking or data breaches.

4.2.1 Test Case for Unit Testing

Unit testing targeted discrete units of code, such as functions, methods, or scripts, to ensure they operate correctly in isolation. This granular approach helped detect early issues in logic, input validation, and database interactions. For instance, tests verified that PHP functions handled edge cases like invalid inputs or concurrent access attempts. Testing was performed in a local XAMPP environment, with assertions checked manually via echo statements for debugging. Key areas included authentication, data retrieval, and CRUD operations.

Table 4.1 User Registration Test Case of OBRS

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Full Name= '' Email ='' Phone No ='' Password=''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Full Name= 'User' Email = 'user@@gmail.com' Phone No = '9876543210' Password= 'Pass123@'	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Full Name= 'User' Email = 'user@gmail.com' Phone No = '9876543210' Password= 'Pass123@'	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass
4	User enters valid details	Full Name= 'Use2r' Email = 'use2r@gmail.com' Phone No = '9876543210' Password= 'Pass123@'	Account created successfully and redirect to login	Account created successfully and redirect to login	Pass

Table 4.2 User Login Test Case of OBRS

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Email ='' Password=''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Email = 'user@@gmail.com' Password='Pass123@'	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Email = 'user@gmail.com' Password='Pass123@'	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass
4	User enters valid details	Email = 'use2r@gmail.com' Password='Pass123@'	Logged in successfully and redirect to user dashboard / admin dashboard	Logged in successfully and redirect to user dashboard / admin dashboard	Pass

Table 4.3 Reservation Test Case of OBRS

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Travel Date = null Seat = '' Passenger Name = '' Contact No = ''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid travel date	Travel Date = 2045-01-01 Seat = '1,2,3' Passenger Name = 'Sangam' Contact No = '9876543210'	Warning Message popup for date validation	Warning Message popup for date validation	Pass

3	User enters invalid phone no	Travel Date = 2026-01-01 Seat = '1,2 ,3' Passenger Name = 'Sangam' Contact No = '09998'	ErrorMessage popup for invalid phone no	ErrorMessage popup for invalid phone no	Pass
4	User enters valid details	Travel Date = 2026-01-01 Seat = '1,2 ,3' Passenger Name = 'Sangam' Contact No = '9876543210'	Reservation Confirmed successfully and redirect to dashboard	Reservation Confirmed successfully and redirect to dashboard	Pass

4.2.2 Test Case for System Testing

System testing evaluates the fully integrated Online Bus Reservation System (OBRS) as a whole, verifying that all modules, user registration, login, bus search, seat reservation, e-ticket generation, and admin management, function correctly and interact seamlessly. The following test cases cover each module of the system, documenting the test scenario, input data, expected output, actual output, and result for each case.

Table 4.4 System Test Case of OBRS

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
TC-01	Register with all valid inputs	Name: Udesb Regmi, Email: udesb@gmail.com, Phone: 9812345678, Password: Test@1234	Account created successfully; redirected to login page	Account created; redirected to login	Pass
TC-02	Register with already existing email	Email: udesb@gmail.com (duplicate)	Error: Email already registered	Error message displayed	Pass

TC-03	Login with valid credentials	Email: sangam@gmail.com, Password: Test@1234	Login successful; redirected to User Dashboard	Dashboard loaded correctly	Pass
TC-04	Login with empty email field	Email: (blank), Password: Test@1234	Validation error: Email is required	Validation error shown	Pass
TC-05	Admin login with admin credentials	Email: admin@bus.com, Password: Admin@123	Login successful; redirected to Admin Dashboard	Admin Dashboard loaded	Pass
TC-06	User attempts to access admin panel directly via URL	URL: /admin/dashboard.php (not logged in as admin)	Access denied; redirected to login page	Redirected to login	Pass
TC-07	Search with valid source and destination	From: Kathmandu, To: Pokhara	List of available buses displayed with schedule and fare	Bus list displayed correctly	Pass
TC-08	Search with valid route but no buses available	From: Kathmandu, To: Mustang (no buses)	Message: No buses available for this route	No buses message shown	Pass
TC-09	Select an available seat	User clicks on a seat (available) seat	Seat highlighted with green as selected; proceed to confirm	Seat selected and highlighted	Pass
TC-10	Attempt to select an already booked seat	User clicks on a red (booked) seat	Seat not selectable; no action taken	Seat unclickable	Pass
TC-11	Attempt to book without	No seat selected; proceed clicked	Validation error: Please select a seat	Validation error shown	Pass

	selecting a seat				
TC-12	View booking history after reservation	User navigates to My Reservations	All past and current bookings listed with status	Booking history displayed	Pass
TC-13	Cancel an existing booking	User selects a booking and clicks Cancel	Booking status updated to Cancelled; seat released	Cancellation processed; seat freed	Pass
TC-14	Complete payment via simulated gateway or cash	User selects Pay Now after booking confirmation	Payment processed; booking status updated to Confirmed	Status updated to Confirmed	Pass
TC-15	E-ticket generated	Completed payment for Booking ID #1045	E-ticket with passenger name, bus, seat, date, and reference number generated	E-ticket generated correctly	Pass
TC-16	Download e-ticket as PDF	User clicks Download Ticket on confirmation page	PDF file downloaded with correct booking details	PDF downloaded successfully	Pass
TC-17	Print e-ticket from browser	User clicks Print Ticket	Printer dialog opens with formatted ticket layout	Print dialog opened	Pass
TC-18	E-ticket displays correct booking information	Booking: Seat 12A, Bus KTM-PKR-01, Date: 2025-11-15	Ticket shows all details: name, bus, route, seat, fare, reference	All details correctly displayed	Pass
TC-19	Booking reference number is	Three separate bookings made by different users	Each booking generates a unique reference number	Unique references generated	Pass

	unique per transaction				
TC-20	Admin adds a new bus with valid details	Bus Number : BA 6 Pa 5919 , Bus Name: Greenline, Route From : KTM, Route To : PKR, Departure Time : 8:00AM, Arrival Time : 12:00PM, Total Seats: 32, Status : Active, Price : NPR 800, Image String: 'url/bus.jpeg'	New bus added; visible in bus list	Bus added successfully	Pass
TC-21	Admin adds bus with empty required fields	Bus Name: (blank), other fields valid	Validation error: Bus name is required	Validation error shown	Pass
TC-22	Admin edits existing bus details	Change fare of Bus ID 3 from NPR 800 to NPR 950	Bus fare updated in database and reflected on search results	Fare updated correctly	Pass
TC-23	Admin deletes a bus with no existing reservations	Delete Bus ID 7 (no bookings)	Bus removed from system; no longer appears in search	Bus deleted successfully	Pass
TC-24	Admin views complete list of all buses	Admin navigates to Manage Bus page	All buses listed with details: name, route, seats, fare, status	Full bus list displayed	Pass
TC-25	Admin views all registered users	Admin navigates to Manage Users page	Complete list of registered users with name,	User list displayed correctly	Pass

			email, phone displayed		
TC-26	Admin views individual user details	Admin clicks on View Details of user Udesb Regmi	Full profile and booking history of selected user shown	User detail page loaded	Pass
TC-27	Admin views all reservations	Admin navigates to Manage Reservations page	All bookings listed with passenger name, bus, seat, status, and date	All reservations displayed	Pass
TC-28	Admin cancels a passenger reservation	Admin selects Booking #1032 and clicks Cancel	Booking status updated to Cancelled; seat released for rebooking	Cancellation processed	Pass
TC-29	Admin views reservation details for a specific booking	Admin clicks on Booking ID #1045	Full booking detail shown: passenger, bus, seat, fare, payment status	Booking detail displayed correctly	Pass
TC-30	Admin dashboard shows summary statistics	Admin logs in and views dashboard	Total users, total bookings, active reservations shown accurately	Dashboard statistics correct	Pass
TC-31	SQL injection attempt in login form	Email: ' OR '1'='1, Password: anything	Login rejected; no unauthorised access granted	Access denied; error shown	Pass
TC-32	Access protected page without login	Navigate to /user/dashboard.php without session	Redirected to login page	Redirected correctly	Pass

TC-33	Password stored as hash in database	New user registers with password Test@1234	Database stores hashed value, not plain text	Hash stored in users table	Pass
TC-34	System accessible on mobile browser	Open OBRS on mobile Brave browser	All pages render responsively without horizontal scroll	Responsive layout confirmed	Pass

Chapter 5: Conclusion & Future Recommendations

5.1. Lesson Learnt Outcome

The development of the Online Based Bus Reservation System (OBRS) has been a valuable learning experience throughout the BCA 4th semester project. Key lessons include mastering the integration of frontend technologies (HTML, CSS, JavaScript, Bootstrap) with backend PHP and MySQL for dynamic web applications, emphasizing the importance of secure coding practices such as prepared statements to prevent SQL injection and session management for authentication. Working within constraints no external APIs, limited to open-source tools, and a two-member team taught efficient time management, modular coding, and the practical application of the Waterfall methodology.

The primary outcome is a fully functional, web-based prototype tailored to Nepal's transportation needs, featuring dual client-admin modules with complete CRUD operations. The system successfully demonstrates real-time bus searching, seat booking, e-ticket generation, and administrative oversight, achieving objectives of reducing manual inefficiencies and enhancing accessibility for passengers across routes serviced by operators.

5.2 Conclusion

In conclusion, the Online Bus Reservation System (OBRS) effectively addresses the persistent challenges in Nepal's traditional bus ticketing processes, such as long queues, overbooking, and limited real-time information, particularly in a geographically diverse nation reliant on road transport. By leveraging accessible open-source technologies PHP for robust backend logic, MySQL for reliable data management, and responsive frontend design the system provides a practical, cost-effective solution that supports both passengers and operators without dependency on external services.

The implementation of client features (route search, seat selection, simulated payments, and e-ticket issuance) alongside comprehensive admin tools (bus management, reservation oversight, and reporting) ensures end-to-end reservation handling. Thorough feasibility analysis, structured design (ER diagrams, DFDs up to Level 1, three-tier architecture), and rigorous testing confirmed the system's technical viability, operational usability, and performance reliability, with near-perfect functionality in simulated environments.

OBRS aligns with Nepal's growing digital infrastructure, complementing existing platforms while offering potential for small-scale operators and public entities like Sajha Yatayat. This project contributes meaningfully to the digitization of public transportation, promoting efficiency, transparency, and inclusivity, and fulfills the academic objectives of the BCA program by demonstrating applied software engineering principles in a real-world Nepali context.

5.3 Future Recommendations:

While the current prototype meets core requirements, several enhancements are recommended for future development to increase scalability, security, and user value:

- **Integration of Live Payment Gateways:** Replace simulated payments with real integrations (e.g., eSewa, Khalti or Mobile Banking) to enable actual transactions, making the system production-ready for commercial use.
- **Mobile Application Development:** Extend the web platform to native Android/iOS apps using frameworks like Flutter or React Native, improving accessibility for Nepal's mobile-first users.
- **Advanced Features:**
 - Implement GPS-based bus tracking and push notifications for delays.
 - Add multilingual support (Nepali, English, and regional languages) and accessibility features for differently-abled users.
 - Incorporate basic analytics or machine learning for demand forecasting and dynamic pricing during peak seasons.
- **Security and Performance Enhancements:** Introduce HTTPS enforcement, CAPTCHA for forms, and load balancing for high-traffic handling during festivals like Dashain.
- **Deployment and Partnerships:** Host on reliable servers and seek collaboration with operators such as Dhading Mandali Yatayat Pvt Ltd, Sakira Travels, Jagadamba Tours and Travels or regional bus associations for real-world pilot testing and adoption.

These improvements would transform OBRS from an academic prototype into a scalable, impactful solution driving further digital transformation in Nepal's public transportation sector.

REFERENCES

- [1] BusSewa Nepal, "Nepal's First Online Real-time Bus Ticket Booking Platform," Diyalo Technologies Pvt. Ltd., Nepal
- [2] Christian, "Online Bus Ticket Reservation Using PHP/MySQLi with Source Code," *Source Codester*, Dec. 2020.
- [3] S. Ghara, T. Saha, A. Ghosh, D. Biswas, and A. Das, "Online Quick Bus Ticket Reservation System," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 10, 2022
- [4] A. Ghimire, A. Paudel, R. Shrestha, and D. Karki, "Online Bus Reservation System," Engineering Project Report, Cosmos College of Management and Technology, Pokhara University, Nepal, Sep. 2022.
- [5] Hamro Bus, "Online Bus Ticketing Platform," Nepal.
- [6] M. O. Oloyede, S. M. Alaya, and K. S. Adewole, "Development of an Online Bus Ticket Reservation System for a Transportation Service in Nigeria," *Computer Engineering and Intelligent Systems*, vol. 5, no. 12, pp. 9–17, 2014.
- [7] J. Saputra, R. Astuti, and W. Prihartono, "Development of a Web-Based Bus Ticket Reservation System for BJM_Pariwisata," *Journal of Artificial Intelligence and Engineering Applications (JAIEA)*, vol. 4, no. 2, pp. 1492–1496, Feb. 2025
- [8] A. K. Singh, "Bus Ticket Booking System," Project Report, Dept. of Computer Science & IT, Arka Jain University, Jharkhand, India, 2023.
- [9] TicketSewa, "Web-Based Bus Reservation and E-Ticketing Solution," G-force Systems & Technologies, Nepal.

APPENDICES

 Nepal Bus Sewa

[Home](#) [View Buses](#) [About](#) [Contact](#) [Admin](#)

Your Journey Starts With Nepal Bus Sewa

Experience reliable, affordable and comfortable travel across the country. Book seats instantly and enjoy stress-free travel with our modern fleet of premium buses.

[Explore Buses](#)

धादिङ मण्डली यातायात प्रा.लि.

काठमाडौं - धादिङ

आरुघाट, सिपकीटार, फूलखर्क, माल्खु
खरी टैनपुर, धुवाकोट, जामलुङ, खाल्टी, मेदी, ग्लाङ

सुरक्षित र भरपर्दो यात्राको लागि



Our Motto

Nepal Bus Sewa is dedicated to redefining road travel through innovation, comfort and dependability. With a strong commitment to safety and quality service, we ensure every journey is smooth, secure and memorable.

Simple Online Booking

Find routes, choose your seat and book in just a few clicks.

Premium Comfort

Spacious seats, AC coaches and modern interiors for a relaxing journey.

Safety First

Experienced drivers, regular bus inspections and real-time monitoring.

Secure Payments

Multiple trusted payment options with encrypted transactions.

Easy Ticketing

Fast and hassle free online ticket booking and management.

Support

24/7 customer support to assist you before, during, and after your trip.

Our Journey

2025

Founded Nepal Bus Sewa with a vision to revolutionize online bus ticketing and travel experience and expanded to 5+ routes nationwide with a fleet of 10+ modern buses

2026

Goal to revolutionizing the bus industry with live payment gateway and AI powered user details & route optimization and premium passenger services.

Ready to Plan Your Journey?

Book your ticket now and enjoy comfortable, affordable travel with our premium bus service across the country.

[Explore Buses Now →](#)

What Our Customers Say



"The seats were super comfortable and the booking system was so easy to use!"
Sital Dangaura



"Great service, friendly staff and the bus arrived exactly on time."
Ranjana Khanal



"Highly reliable travel partner. I use them every month for my work trips."
Sabitri Pandit

Nepal Bus Sewa
Reliable travel. Simple booking.

© 2026 Nepal Bus Sewa. All rights reserved.
Developed By Udesb Regmi & Sangam Sedai

[Home](#) [About](#) [Contact](#) [View Bus](#) [Login](#) [Register](#)

Home Page

38

Nepal Bus Sewa

[Home](#)
[View Buses](#)
[About](#)
[Contact](#)
[Admin](#)

Admin Panel

Dashboard

Manage Buses

Reservations

Users

Forgot Password

Make Reservation

Simrik AC

Ba Pr -01- 06 Kha - 7176

Katmandu → Kakarvita

05:00 AM

Rs. 2,000.00 per seat

Travel Date

14/03/2026

Simrik AC - Ba Pr -01- 06 Kha - 7176

40 Seats Total

0 Reserved, 40 Available

Choose Your Seats (Multiple Selection Allowed)

Selected seats: 2, 3, 6, 11

Click seats to select/deselect. You can select multiple seats.

Available

Selected

Reserved

Window

Aisle

Passenger Name

Admin

Contact Number

9876543210

Payment Method

Cash Payment

Pay at counter

eSewa

Pay online

Total Amount: Rs. 8000.00

Number of seats: 4

Confirm Reservation

Nepal Bus Sewa

Reliable travel. Simple booking.

© 2026 Nepal Bus Sewa. All rights reserved.

Developed By Uddesh Regmi & Sangam Sedai

Home

About

Contact

View Bus

Login

Register

Reserve Bus Page

39

Nepal Bus Sewa

Home

View Buses

About

Contact

Admin

Admin Panel

Dashboard

Manage Buses

Reservations

Users

Forgot Password

Admin Dashboard

8

Total Buses

33

Total Reservations

7

Total Users

Recent Reservations

ID	Passenger	Passenger Phone	Passenger Email	Bus	Date	Amount	Payment Method	Status
52	Sangam Sedai	9876543210	sangamsedai@bus.com	King Air Bus	Mar 14, 2026	Rs. 1,699.00	cash	Confirmed
50	Admin	9876543210	admin@bus.com	Pachpokhari Air Bus	Feb 26, 2026	Rs. 1,999.00	cash	Confirmed
49	Admin	9876543210	admin@bus.com	Pachpokhari Air Bus	Feb 26, 2026	Rs. 1,999.00	cash	Cancelled
51	Admin	9876543210	admin@bus.com	Pachpokhari Air Bus	Feb 26, 2026	Rs. 1,999.00	cash	Confirmed
43	Admin	9838943894	admin@bus.com	Pachpokhari Air Bus	Feb 26, 2026	Rs. 1,999.00	cash	Cancelled

Admin Dashboard of OBRS

Nepal Bus Sewa

View Buses

About

Contact

Sangam Sedai

User Panel

Dashboard

Search Buses

My Reservations

Profile

Welcome, Sangam Sedai!

6

Total Reservations

5

Active Bookings

Recent Reservations

Bus No:	Bus	Route	Date	Seat No.	Status
Su Pr - 02 - 006 Kha 4444	King Air Bus	Dang → Kathmandu	Mar 14, 2026	29	Confirmed
Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	14	Confirmed
Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	15	Confirmed
Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	19	Confirmed
Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	18	Confirmed

User Dashboard for OBRS

40